SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING **Name:** T. Thanmai Reddy **COURSE CODE:** CSA4303 **Reg. No.:** 192425372

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code-along exercise: Make your first HTML page together with CSS **Weightage:** 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>

B. <video src="video.mp4" controls></video>

C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET B. POST C. PUT D. DELETE

Snippet 3: Navigation Elements

A.

B. <nav>

C. <section>

D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>

B. <note><to>User</to><from>Admin</from></note>

C. <note><to>User</from></note>

D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP B. HTTP C. SMTP D. SNMP

Questions

1. Select the correct HTML tag for video embedding and justify your answer.
2. Identify the appropriate HTTP method for form submission and explain why.
3. Choose the proper semantic element for navigation and justify its usage.
4. Identify the correct XML syntax and explain the error in other options.
5. Choose the correct protocol for web communication and justify your choice.

Code of Conduct

I __T. Thanmai Reddy / 192425372_____ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ___T. Thanmai Reddy __

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	17/25	Thorough understanding; clearly identifies all key issues.	Good understanding with most issues identified.	Basic understanding; some issues missed.	Poor understanding; problem not identified correctly.
Application of Concepts	21/25	Effectively applies relevant concepts to analyze the case deeply.	Applies appropriate concepts with minor gaps.	Limited application of concepts.	Fails to apply relevant concepts.
Analysis	16/20	In-depth analysis with strong reasoning and insights.	Good analysis with logical reasoning.	Basic analysis with limited depth.	Weak or no analysis; lacks reasoning.
Solution Design	19/20	Innovative, feasible solutions with strong justification.	Logical solutions with adequate justification.	Basic solutions with weak justification.	Incorrect or no solution provided.
Clarity	9/10	Clear, well structured, and easy to understand.	Organized and understandable presentation.	Limited clarity and structure.	Poor clarity; difficult to understand.

Total:82/100

1. Select the Correct HTML Tag for Video Embedding

Correct Answer: Option B — <video src="video.mp4" controls></video>

The HTML5 <video> element embeds a video in a webpage. The src attribute identifies the video file, and controls displays browser-provided playback controls.

```
<video src="video.mp4" controls width="640">
 Your browser does not support the video element.
</video>
```

Why the other options are incorrect:

- <media> is not the standard HTML element for embedding video.
- <movie> is not a standard HTML element.
- <vid> is not a standard HTML element.

The video file must exist at the specified path. Provide captions when appropriate and use a supported video format.

HTML code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>HTML5 Video Embedding</title>

 <style>
* {
box-sizing: border-box;
}

body {
margin: 0;
font-family: Arial, sans-serif;
background: #eef4f8;
color: #243447;
}

header {
```

```
background: #154360;
color: white;
text-align: center;
padding: 25px;
}

main {
max-width: 850px;
margin: 30px auto;
padding: 25px;
background: white;
border-radius: 12px;
box-shadow: 0 4px 14px #cbd5df;  }

h2 {
color: #154360;
}

pre {
background: #17202a;
color: #a9dfbf;
padding: 16px;
border-radius: 8px;
overflow-x: auto;
}

video {
display: block;
max-width: 100%;
margin: 15px 0;
background: #111;
}

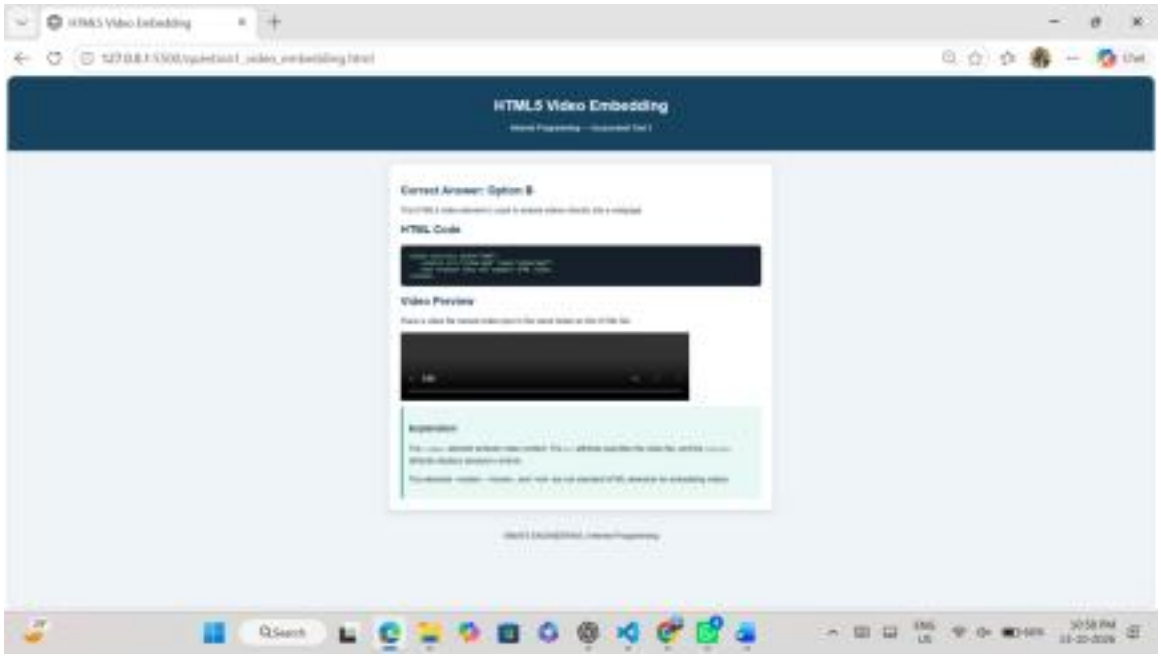
.note {
background: #e8f8f5;
padding: 14px;
border-left: 5px solid #148f77;
```

```
 border-radius: 5px;
 line-height: 1.6;
 }

 footer {
 text-align: center;
 padding: 18px;
 }
 </style>
</head>
<body>
<header>
 <h1>HTML5 Video Embedding</h1>
 <p>Internet Programming — Assessment Tool 3</p>
</header>
<main>
 <h2>Correct Answer: Option B</h2>
 <p>
 The HTML5 video element is used to embed videos
directly into a webpage.
 </p>
 <h2>HTML Code</h2>
 <pre><code>&lt;video controls width="640"&gt;
&lt;source src="video.mp4" type="video/mp4"&gt;
Your browser does not support HTML video.
&lt;/video&gt;</code></pre>
 <h2>Video Preview</h2>
 <p>
 Place a video file named video.mp4 in the same
folder as this HTML file.
 </p>
 <video controls width="640">
 <source src="video.mp4" type="video/mp4">
Your browser does not support HTML video.
</video>
 <div class="note">
 <h3>Explanation</h3>
```

```
 <p>
 The <code>&lt;video&gt;</code> element embeds video content.  The
<code>src</code> attribute specifies the video file,
 and the <code>controls</code> attribute displays
 playback controls.
 </p>
 <p>
 The elements &lt;media&gt;, &lt;movie&gt;, and
 &lt;vid&gt; are not standard HTML elements
 for embedding videos.
 </p>
 </div>
</main>
<footer>
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</footer>
</body>
</html>
```

Output: -



2. Identify the Appropriate HTTP Method for Form Submission **Correct Answer: Option B** — POST

POST is appropriate for submitting registration data because form values are sent in the request body rather than appended to the URL. POST does not encrypt the data; HTTPS and server-side validation are still required.

```
<form action="/register" method="POST">
 <label for="name">Name:</label>
 <input type="text" id="name" name="name" required>
 <label for="email">Email:</label>
 <input type="email" id="email" name="email" required>
 <button type="submit">Register</button>
</form>
```

HTML code:-

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Registration</title>
 <style>
 * {
 box-sizing: border-box;
 }
 body {
 margin: 0;
 font-family: Arial, sans-serif;
 background: #e8f6f3;
 color: #263238;
 }
 header {
 background: #117864;
 color: white;
 text-align: center;
 padding: 25px;
 }
 main {
 max-width: 560px;
 margin: 30px auto;
 padding: 26px;
 background: white;
 border-radius: 12px;
```

```
box-shadow: 0 4px 14px #b7d8d0;  }
h2 {
color: #117864;
}
label {
display: block;
margin: 14px 0 6px;
}
input {
width: 100%;
padding: 11px;
border: 1px solid #aaa;
border-radius: 5px;
}
button {
margin-top: 20px;
padding: 12px 18px;
background: #117864;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}
button:hover {
background: #0b5345;
}
.notice {
margin-top: 16px;
padding: 14px;
border-radius: 6px;
background: #fcf3cf;
line-height: 1.6;
}
.result {
margin-top: 16px;
padding: 14px;
border-radius: 6px;
```

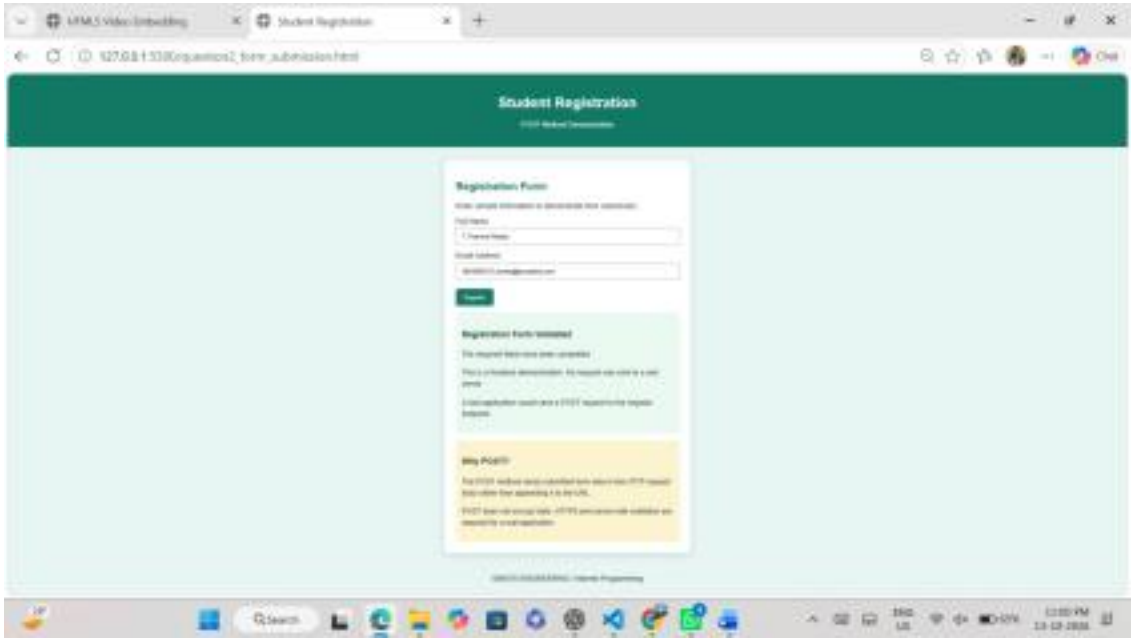
```
background: #eafaf1;
line-height: 1.6;
}
footer {
text-align: center;
padding: 18px;
}
</style>
</head>
<body>
<header>
<h1>Student Registration</h1>
<p>POST Method Demonstration</p>
</header>
<main>
<h2>Registration Form</h2>
<p>
Enter sample information to demonstrate form submission.
</p>
<form id="registrationForm"
action="/register"
method="POST">
<label for="name">Full Name</label>
<input
id="name"
name="name"
type="text"
required
<label for="email">Email Address</label>

<input
id="email"
name="email"
type="email"
required
>
<button type="submit">
```

```
 Register
 </button>
 </form>
 <div id="result" class="result" hidden></div>
<div class="notice">
 <h3>Why POST?</h3>
 <p>
 The POST method sends submitted form data  in
the HTTP request body rather than  appending it to
the URL.
 </p>
 <p>
 POST does not encrypt data. HTTPS and
server-side validation are required  for a real
application.
 </p>
 </div>
</main>
<footer>
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</footer>
<script>
 document
 .getElementById("registrationForm")
.addEventListener("submit", function(event) {
event.preventDefault();
 const result =
 document.getElementById("result");
 result.hidden = false;
 result.innerHTML = `
 <h3>Registration Form Validated</h3>
 <p>
 The required fields have been completed.
 </p>
 <p>
 This is a frontend demonstration.
 No request was sent to a real server.
```

```
</p>
<p>
A real application would send a POST
request to the /register endpoint.
</p>
`;
});
</script>
</body>
</html>
```

Output: -



3. Choose the Proper Semantic Element for Navigation

Correct Answer: Option B — <nav>

The <nav> element identifies a section containing major navigation links. It provides semantic meaning and helps browsers, developers, and assistive technologies understand the purpose of the links.

```
<nav aria-label="Main navigation">
 <a href="index.html">Home</a>
 <a href="about.html">About</a>
 <a href="services.html">Services</a>
 <a href="contact.html">Contact</a>
</nav>
```

A is a generic container, <section> groups related content, and <header> represents

introductory content. These do not replace <nav> when the purpose is a navigation menu.

HTML Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Semantic Navigation</title>
 <style>
 * {
 box-sizing: border-box;
 }
 body {
 margin: 0;
 font-family: Arial, sans-serif;
 background: #f4ecf7;
 color: #333;
 }
 header {
 background: #6c3483;
 color: white;
 text-align: center;
 padding: 26px;
 }
 nav {
 display: flex;
gap: 8px;
 justify-content: center;
 flex-wrap: wrap;
 background: #512e5f;
 padding: 14px;
 }
 nav a {
 color: white;
 text-decoration: none;
 padding: 10px 15px;
```

```
border-radius: 5px;
}
nav a:hover,
nav a:focus {
background: #8e44ad;
outline: 2px solid white;  }
main {
max-width: 850px;
margin: 30px auto;
padding: 25px;
background: white;
border-radius: 12px;
box-shadow: 0 4px 14px #d7bde2;  }
h2 {
color: #6c3483;
}
pre {
background: #f4f4f4;
padding: 16px;
border-radius: 6px;
overflow-x: auto;
}
section {
padding: 10px 0;
}
footer {
text-align: center;
padding: 18px;
}
</style>
</head>
<body>
<header>
 <h1>College Web Portal</h1>
 <p>Semantic HTML Navigation Example</p>
</header>
<nav aria-label="Main navigation">
```

Home
About
Courses
Contact
</nav>
<main id="home">
<section id="about">
<h2>About Us</h2>

Welcome to our college web portal. This page demonstrates semantic navigation using HTML5.

</section>
<section id="courses">
<h2>Courses</h2>

Our portal provides information about courses, academic activities, and events.
</section>
<section id="contact">
<h2>Contact</h2>

Email: info@example.com

</section>
<h2>HTML Code</h2>
<pre><code><nav aria-label="Main navigation">
Home
About
Courses
Contact
</nav></code></pre>
<h2>Explanation</h2>

The <code><nav></code> element identifies a section containing major navigation links.

The <code><div></code> element is a generic container, whereas <code><nav></code> provides semantic meaning.

</p>

</main>

<footer>

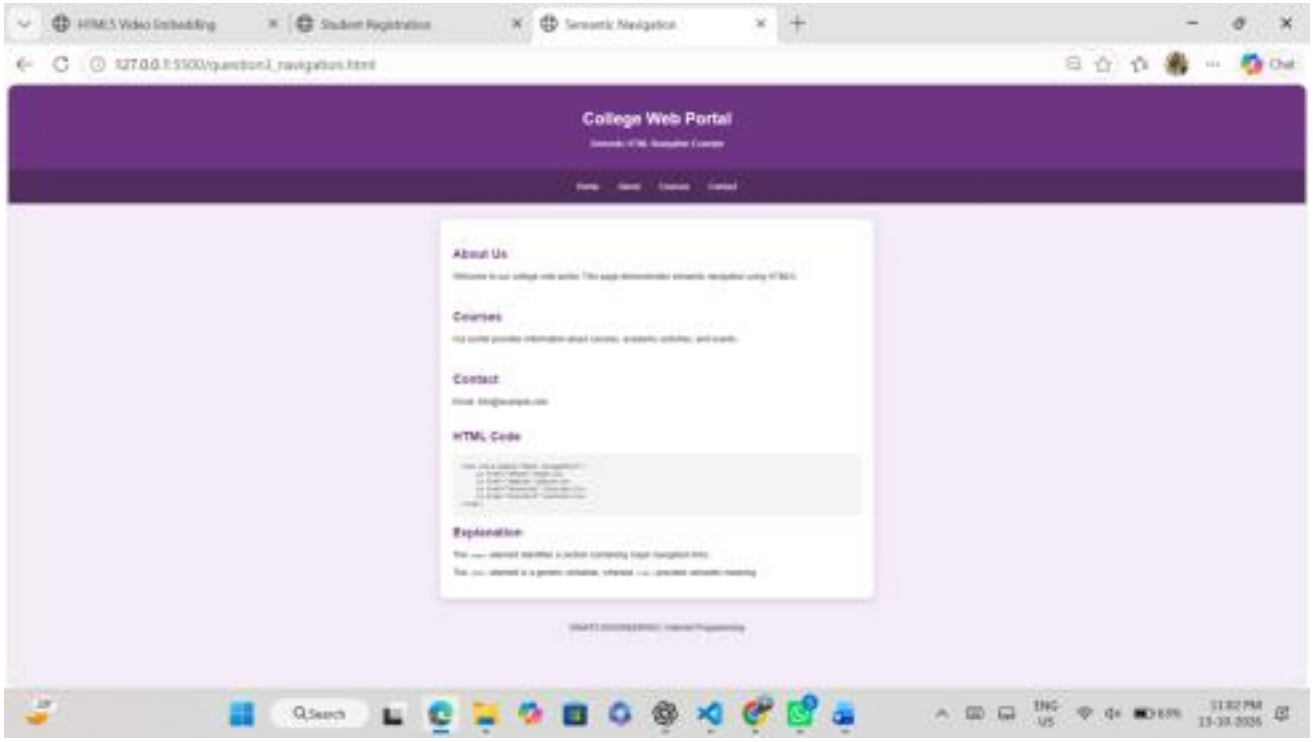
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</footer>

</body>

</html>

Output: -



4. Identify the Correct XML Syntax

Correct Answer: Option B — <note><to>User</to><from>Admin</from></note> XML elements must be properly nested and closed, and opening and closing tag names must match.

```
<note>
 <to>User</to>
 <from>Admin</from>
</note>
```

Errors in the other options:

- Option A: The root <note> element is not closed.
- Option C: The <to> element is incorrectly closed with </from>, so the tags do not match. •

Option D: The final angle bracket is missing; a valid form is <note to="User"></note> or <note to="User"/>.

XML parsers generally reject malformed XML instead of applying HTML-style error recovery.

HTML Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>XML Syntax Checker</title>
 <style>
 * {
 box-sizing: border-box;
 }
 body {
 margin: 0;
 font-family: Arial, sans-serif;
 background: #fef5e7;
 color: #333;
 }
 header {
 background: #b9770e;
 color: white;
 text-align: center;
 padding: 24px;
 }
 main {
 max-width: 850px;
 margin: 30px auto;
 padding: 25px;
 background: white;
 border-radius: 12px;
 box-shadow: 0 4px 14px #e6d1ad;
 }
 h2 {
```

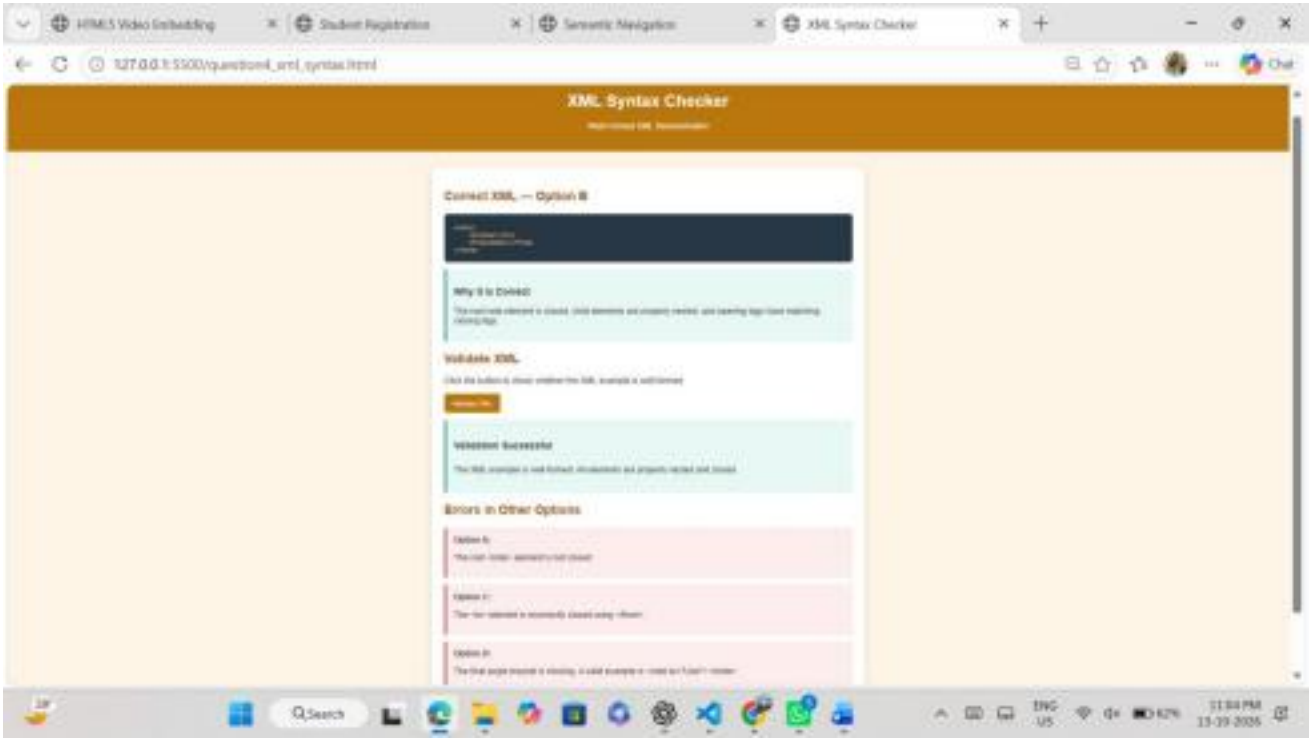
```
 color: #935116;
 }
 pre {
 background: #273746;
 color: #f8c471;
 padding: 18px;
 border-radius: 7px;
 overflow-x: auto;
 }
 .good {
 background: #e8f8f5;  border-left:
5px solid #148f77;  padding: 15px;
 border-radius: 5px;
 }
 .bad {
 background: #fdedec;  border-left:
5px solid #c0392b;  padding: 15px;
 border-radius: 5px;
 margin-top: 12px;
 }
 button {
 padding: 11px 16px;
 background: #b9770e;
color: white;
 border: none;
 border-radius: 5px;
 cursor: pointer;
 }
 button:hover {
 background: #784212;  }
 #validation {
 margin-top: 15px;
 line-height: 1.7;
 }
 footer {
 text-align: center;
 padding: 18px;
```

```
 }
 </style>
</head>
<body>
<header>
 <h1>XML Syntax Checker</h1>
 <p>Well-Formed XML Demonstration</p>
</header>
<main>
 <h2>Correct XML — Option B</h2>
 <pre><code>&lt;note&gt;
 &lt;to&gt;User&lt;/to&gt;
 &lt;from&gt;Admin&lt;/from&gt;
&lt;/note&gt;</code></pre>
 <div class="good">
 <h3>Why It Is Correct</h3>
 <p>
 The root note element is closed, child elements are
properly nested, and opening tags have matching
closing tags.
 </p>
 </div>
 <h2>Validate XML</h2>
 <p>
 Click the button to check whether the XML example is
well-formed.
 </p>
 <button onclick="validateXML()">
 Validate XML
 </button>
 <div id="validation" aria-live="polite"></div>
<h2>Errors in Other Options</h2>
 <div class="bad">
 <b>Option A:</b>
 <p>
 The root &lt;note&gt; element is not closed. </p>
 </div>
```

```
 <div class="bad">
 <b>Option C:</b>
 <p>
 The &lt;to&gt; element is incorrectly closed  using
&lt;/from&gt;.
 </p>
 </div>
 <div class="bad">
 <b>Option D:</b>
 <p>
 The final angle bracket is missing.
 A valid example is
 &lt;note to="User"&gt;&lt;/note&gt;.
 </p>
 </div>
</main>
<footer>
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</footer>
<script>
 function validateXML() {
 const xmlText =
 '<note><to>User</to><from>Admin</from></note>';  const
parsed =
 new DOMParser().parseFromString(
 xmlText,
 "application/xml"
 );
 const result =
 document.getElementById("validation");  if
(parsed.querySelector("parsererror")) {
 result.innerHTML = `
 <div class="bad">
 XML is not well-formed.
 </div>
 `;
 } else {
```

```
result.innerHTML = `
<div class="good">
<h3>Validation Successful</h3>
<p>
The XML example is well-formed.
All elements are properly nested
and closed.
</p>
</div>
`;
}
}
</script>
</body>
</html>
```

Output: -



5. Choose the Correct Protocol for Web Communication

Correct Answer: Option B — HTTP

HTTP (Hypertext Transfer Protocol) is used for communication between web clients, such as browsers, and web servers. It defines how requests and responses are exchanged. HTTPS is HTTP carried over TLS, providing encryption and server authentication.

Why the other options are incorrect:

- FTP is used for file transfer.
- SMTP is used to send email.
- SNMP is used for network management and monitoring.

```
Browser -- HTTP/HTTPS Request --> Web Server
Browser <-- HTTP/HTTPS Response -- Web Server
```

For public websites and registration forms, HTTPS should be used to protect data in transit. **HTML Code: -**

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Web Communication Protocols</title>
 <style>
 * {
 box-sizing: border-box;
 }
 body {
 margin: 0;
 font-family: Arial, sans-serif;
 background: #ebf5fb;
 color: #2c3e50;
 }
 header {
 background: #1f618d;
 color: white;
 text-align: center;
 padding: 24px;
 }
 main {
```

```
max-width: 850px;
margin: 30px auto;
padding: 25px;
background: white;
border-radius: 12px;
box-shadow: 0 4px 14px #aed6f1;  }
h2 {
color: #1f618d;
}
.flow {
display: grid;
grid-template-columns: 1fr auto 1fr;  gap:
12px;
align-items: center;
margin: 20px 0;
}
.box {
padding: 18px;
text-align: center;
background: #d6eaf8;
border-radius: 8px;
border: 1px solid #85c1e9;  }
.arrow {
text-align: center;
font-size: 24px;
color: #1f618d;
}
button {
padding: 12px 18px;
background: #1f618d;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}
button:hover {
background: #154360;  }
```

```
 #result {
 margin-top: 16px;
 padding: 15px;
 background: #e8f8f5;  border-left:
5px solid #148f77;  border-radius:
5px;
 line-height: 1.7;
 }
 pre {
 background: #273746;
color: #d6eaf8;
 padding: 16px;
 border-radius: 6px;
 overflow-x: auto;
 }
 footer {
 text-align: center;
 padding: 18px;
 }
 @media (max-width: 600px) {
.flow {
 grid-template-columns: 1fr;  }
 .arrow {
 transform: rotate(90deg);  }
 }
 </style>
</head>
<body>
<header>
 <h1>Web Communication Protocols</h1>
<p>HTTP Request and Response</p>
</header>
<main>
 <h2>Browser–Server Communication</h2>
<div class="flow">
 <div class="box">
 <h3>Web Browser</h3>
```

Creates a request

⇄

<h3>Web Server</h3>
Processes the request

<button onclick="showDetails()">
Show HTTP Communication
</button>
<div id="result" hidden>
<h2>Example HTTP Request and Response</h2>
<pre><code>GET /index.html HTTP/1.1 Host:
example.com
HTTP/1.1 200 OK
Content-Type: text/html</code></pre>
<h2>Protocol Comparison</h2>

HTTP:
Transfers web requests and responses.

HTTPS:
HTTP protected with TLS encryption.

FTP:
Used for file transfer.

SMTP:
Used to send email.

SNMP:
Used for network management and monitoring.


```
 <h2>Correct Answer: Option B — HTTP</h2>
<p>
 HTTP stands for Hypertext Transfer Protocol.  It
defines how browsers and web servers exchange
requests and responses.
 </p>
 <p>
 HTTPS should be used for real websites to protect
communication using TLS.
 </p>
</main>
<footer>
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</footer>
<script>
 function showDetails() {
 const result =
 document.getElementById("result");
result.hidden = false;
 result.innerHTML = `
 <h3>HTTP Communication Process</h3>  <p>
 <b>Step 1:</b>
 The browser prepares an HTTP request.
 </p>
 <p>
 <b>Step 2:</b>
 The request travels to the web server.  </p>
 <p>
 <b>Step 3:</b>
 The server processes the request and
prepares a response.
 </p>
 <p>
 <b>Step 4:</b>
 The server returns a status code,
headers, and response content.  </p>
 <p>
```

```
<b>Step 5:</b>
The browser interprets the response  and
displays the webpage.  </p>
`;
result.style.cssText =
"margin-top:16px;padding:15px;" +
"background:#e8f8f5;" +
"border-left:5px solid #148f77;" +
"border-radius:5px;line-height:1.7";  }
</script>
</body>
</html>
```

Output: -

